# HSE 103 Final Project: Submission 1

Azaire Andre

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## Data Import

```
# Read in covariate data
covarData <- read.csv(file = "GSE85307_covar-1.csv")

# Read in gene expression data
geneExpression <- read.csv(file = "GSE85307_geneExpression.csv",
                           row.names = 1)

# Read in gene name linker
geneNames <- read.csv(file = "GSE85307_geneNames.csv",
                      row.names = 1)

# Check dimensions
dim(covarData)
```

```
## [1] 157  10
```

```
dim(geneExpression)
```

```
## [1] 500 157
```

```
dim(geneNames)
```

```
## [1] 750   1
```

```
# Check if geoID values match gene expression column names
all(covarData$geoID %in% colnames(geneExpression))
```

```
## [1] TRUE
```

```
# Check if all gene expression column names are in covariate data
all(colnames(geneExpression) %in% covarData$geoID)
```

```
## [1] TRUE
```

```r
# Find probe ID for TBL1X
geneNames[geneNames[,1] == "TBL1X", , drop = FALSE]
```

```
##          GENE
## 8165911 TBL1X
```

```r
# Create project data frame
projectData <- covarData

# Add TBL1X expression matched by geoID
projectData$TBL1X <- as.numeric(unlist(
  geneExpression["8165911", projectData$geoID]
))

# Check TBL1X values
summary(projectData$TBL1X)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   8.283   8.955   9.194   9.165   9.421   9.862
```

```r
# Remove text from BMI values
projectData$mat_BMI <- sub("bmi_kg/m2: ",
                           "",
                           projectData$mat_BMI)

# Convert BMI to numeric
projectData$mat_BMI <- as.numeric(projectData$mat_BMI)

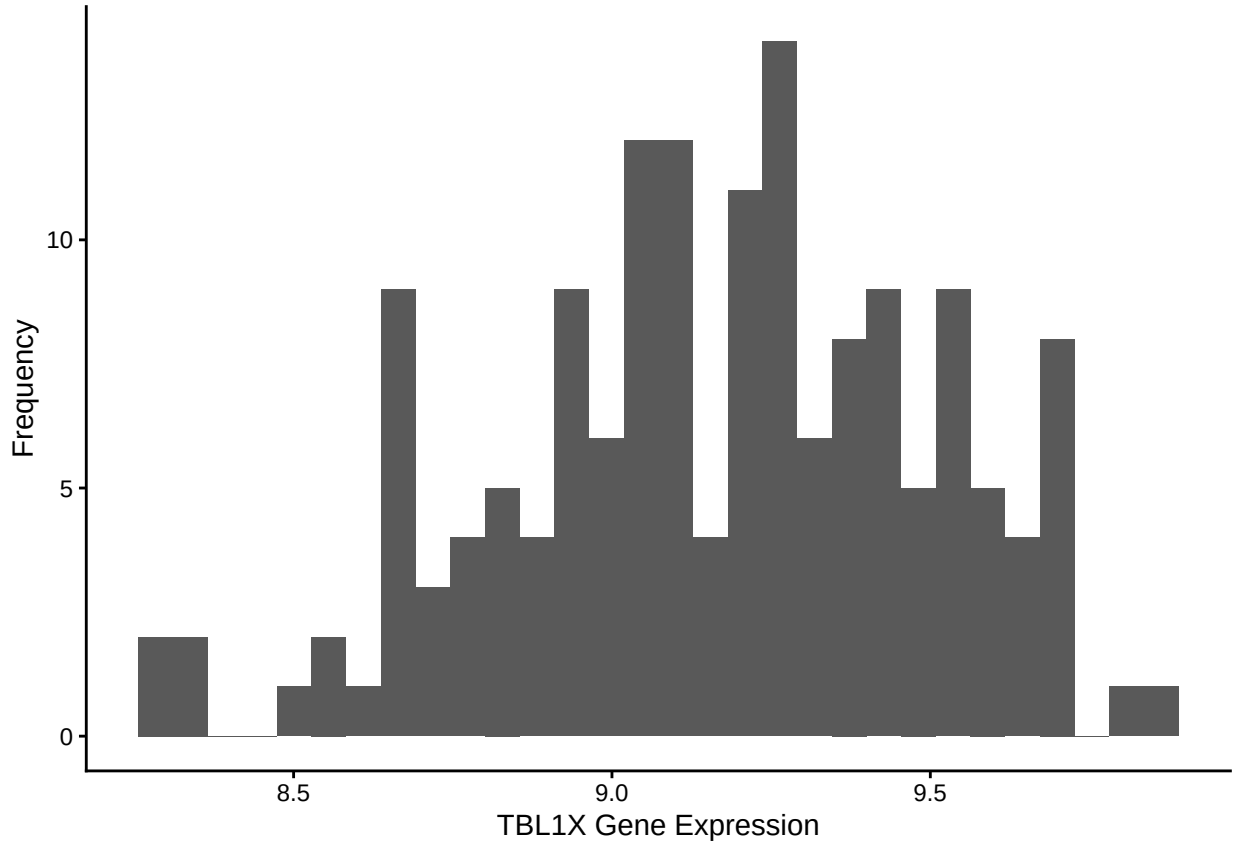
# Calculate median BMI
medianBMI <- median(projectData$mat_BMI,
                    na.rm = TRUE)

# Create categorical BMI variable
projectData$BMIgroup <- cut(projectData$mat_BMI,
                            breaks = c(-Inf, medianBMI, Inf),
                            labels = c('Lower BMI','Higher BMI'))

# Check number of subjects in each BMI group
table(projectData$BMIgroup)
```

```
##
##  Lower BMI Higher BMI
##         79         78
```

```r
# Generate histogram of TBL1X expression
ggplot(projectData,
       aes(x = TBL1X)) +
  geom_histogram(bins = 30) +
  labs(x = 'TBL1X Gene Expression',
       y = 'Frequency') +
  theme_classic()
```




```
# Restore original vitamin D values
projectData$mat_vitD <- covarData$mat_vitD

# Remove text from vitamin D values
projectData$mat_vitD <- sub("vitamind_25\\(0h\\)_ng/ml_whole_blood_baseline: ",
                            "",
                            projectData$mat_vitD)

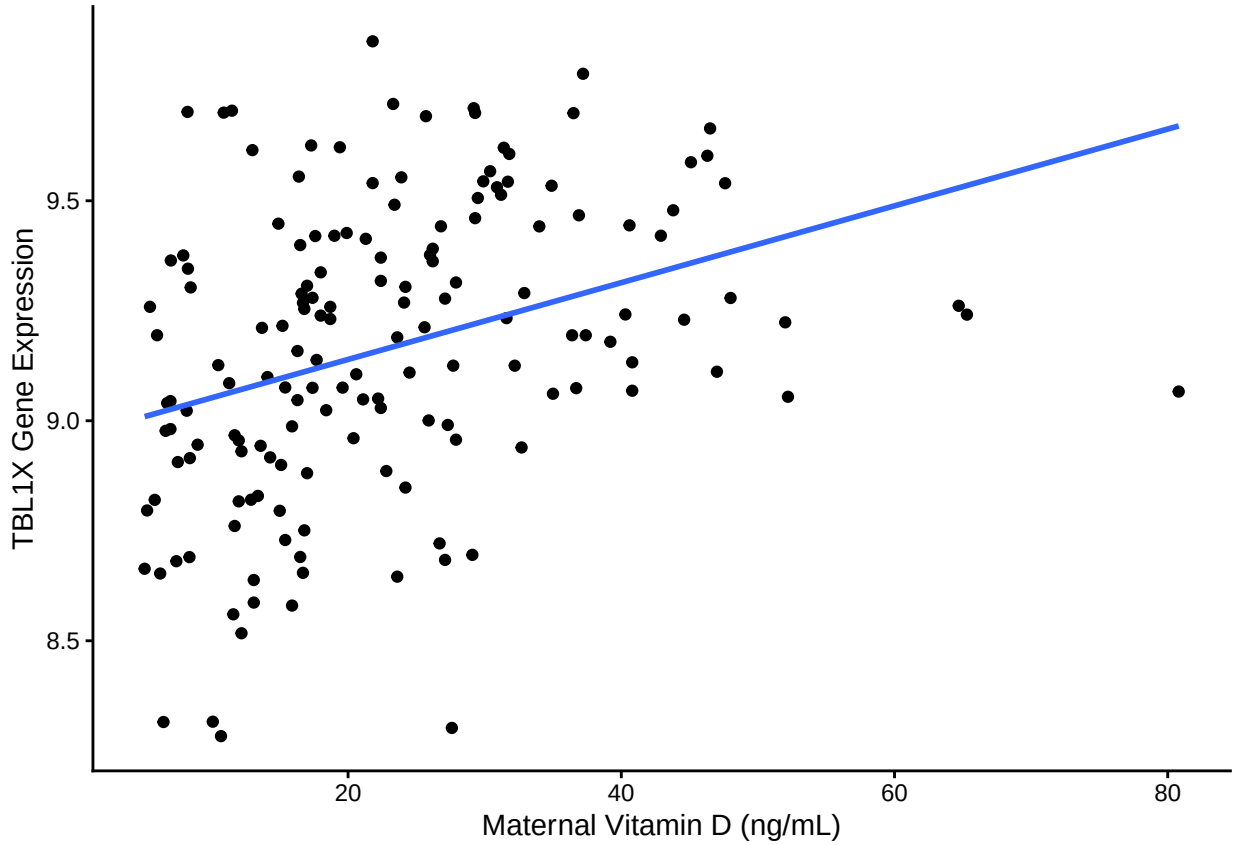
# Convert vitamin D to numeric
projectData$mat_vitD <- as.numeric(projectData$mat_vitD)

# Check vitamin D values
summary(projectData$mat_vitD)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5.11   13.40   20.40   22.98   29.50   80.80
```

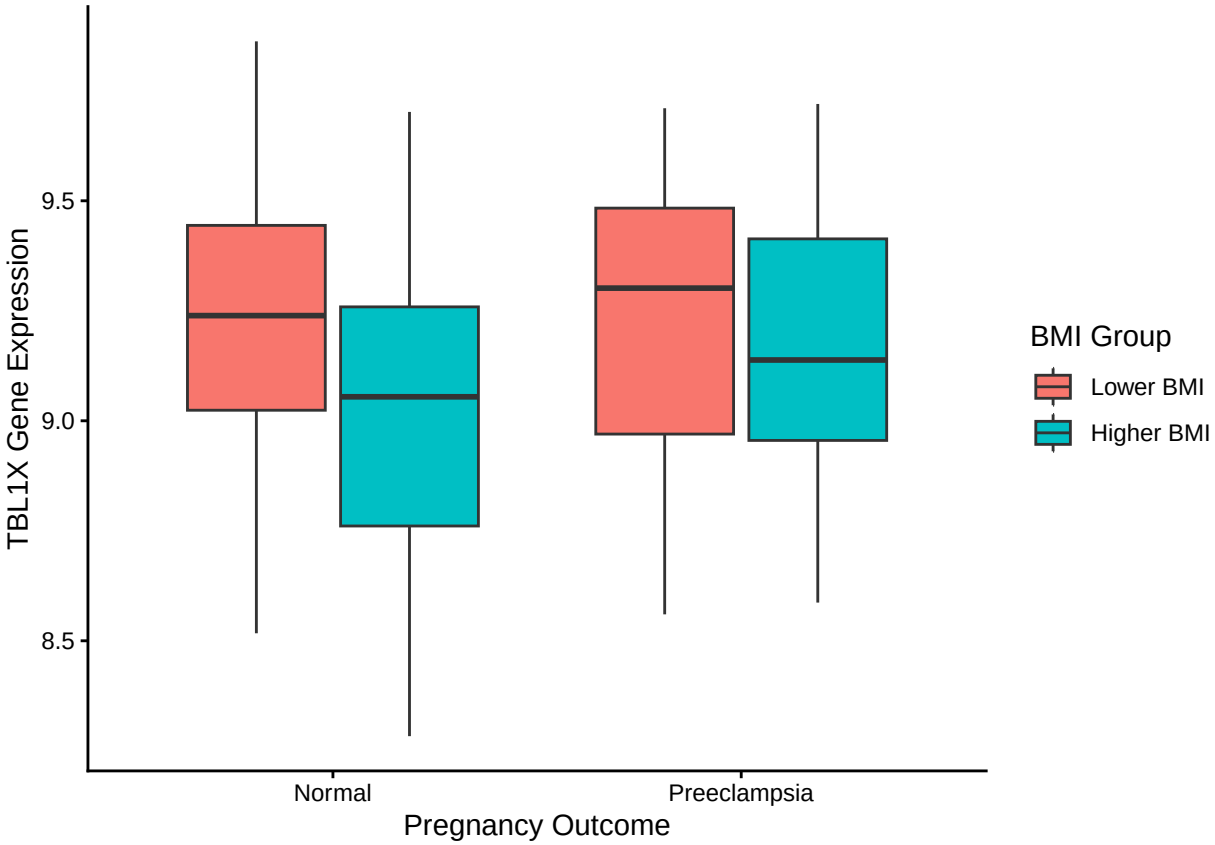
```
# Generate scatterplot
ggplot(projectData,
       aes(x = mat_vitD,
           y = TBL1X)) +
  geom_point() +
  geom_smooth(method = lm,
              se = FALSE) +
  labs(x = 'Maternal Vitamin D (ng/mL)',
```

```
       y = 'TBL1X Gene Expression') +
  theme_classic()
```




```
# Remove text from preeclampsia values
projectData$preg_preeclampsia <- sub("pregnancy_condition: ",
                                     "",
                                     projectData$preg_preeclampsia)

# Generate boxplot
ggplot(projectData,
       aes(x = preg_preeclampsia,
           y = TBL1X,
           fill = BMIgroup)) +
  geom_boxplot() +
  labs(x = 'Pregnancy Outcome',
       y = 'TBL1X Gene Expression',
       fill = 'BMI Group') +
  theme_classic()
```


TBL1X Gene Expression
9.5
9.0
8.5
Normal
Preeclampsia
Pregnancy Outcome
BMI Group
Lower BMI
Higher BMI